ISTE • AVIXA

PROFESSIONAL DEVELOPMENT INITIATIVE

Classroom **AV Essentials**

Day 1 — AV Fundamentals

ISTE × AVIXA | PROFESSIONAL DEVELOPMENT INITIATIVE

"Spend less time troubleshooting.
Spend more time teaching."

AV FUNDAMENTALS · DAY 1 · PROFESSIONAL D

DAY 1 OF 2 | AV ECOSYSTEM · AUDIO · VIDEO · SIGNAL FLOW · NETWORKING · ACCESSIBILITY

7 HOURS · FULL DAY

DAY 1 — JUNE 2026

Today at a Glance

Instructional Module
Break / Lunch
7 Hours Total

8:00 – 8:30	Welcome & Introductions	30 min
8:30 – 9:30	Module 1	60 MIN
9:30 – 9:45	Break	15 MIN
9:45 – 11:00	Module 2	75 MIN
11:00 – 12:00	Module 3	60 MIN
12:00 – 1:00	Lunch	60 MIN
1:00 – 2:00	Module 4	60 MIN
2:00 – 2:15	Break	15 MIN
2:15 – 3:15	Module 5	60 MIN
3:15 – 4:00	Module 6	45 MIN
4:00 – 4:30	Day 1 Wrap-Up + Kit Review	30 min

6 modules across 4 content hours — all materials in your handbook. Breaks are firm.

You Don't Need to Be an **AV Technician.**

5 min · Ask: "Worst AV moment?"

Understand

Know what the system is **supposed to do** — and what normal looks, sounds, and feels like before anything goes wrong.

LAYER 1 OF 3

Troubleshoot

Know how to **diagnose common failures in 5 minutes** — before calling IT, and without losing your class momentum.

LAYER 2 OF 3

Teach

Keep **instruction moving when things go wrong** — because your students need you present, not panicked at a cable panel.

LAYER 3 OF 3

Today is about building AV confidence — not AV expertise.

MODULE 01

The AV Ecosystem

Understanding the Four Layers of Your Classroom System

60 minutes | Lecture + Discussion

LAYER 1
Source Devices

LAYER 2
Display System

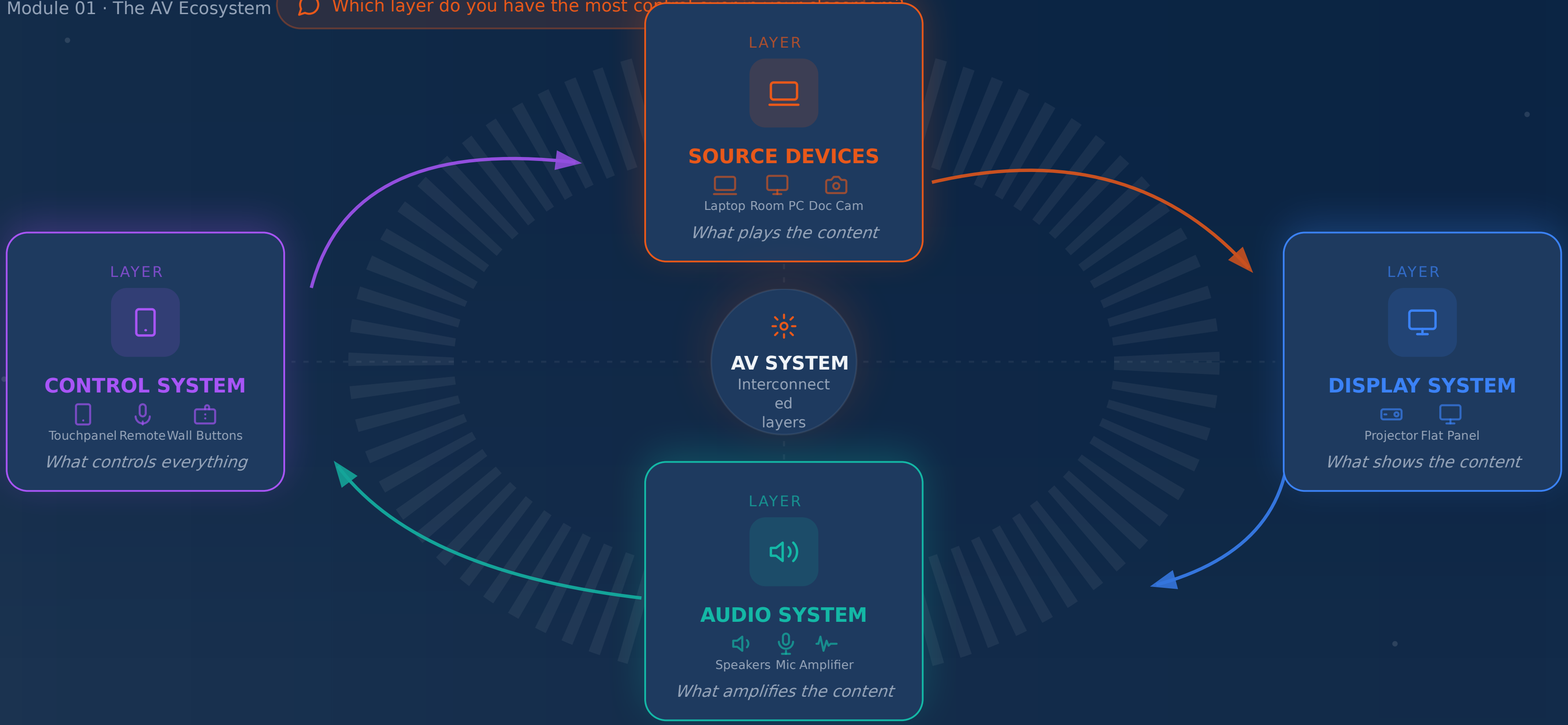
LAYER 3
Audio System

LAYER 4
Control System

The Four Layers of Your Classroom AV System

Module 01 · The AV Ecosystem

Which layer do you have the most co…



min · Lecture + Discussion

Source Display Audio Control

MODULE 01

Who Owns What — And What Can You Control?

Know your scope of control in each layer of the classroom AV system

CONTROL LEVEL
- HIGH
- MEDIUM
- LOW–MEDIUM

LAYER	EXAMPLES	WHO CONTROLS IT	YOUR LEVEL OF CONTROL
Source Devices Layer 1	Laptop, Room PC Document Camera		HIGH You control these fully
Display System Layer 2	Projector, Flat Panel Projection Screen		MEDIUM Power, brightness, input
Audio System Layer 3	Speakers, Mic, Amplifier DSP Processor		LOW–MEDIUM Volume, mute, mic selection
Control System Layer 4	Touchpanel, Remote Wall Buttons		MEDIUM Interface only — not configuration

Speaker note (5 min): Ask — "Which layer do you actually touch every day? Which one have you never changed?"

⚠ "Teacher fully controls only Layer 1 — everything else requires coordinatio

7 min | Lecture

Key AV Terms — Plain English

TERM		PLAIN-ENGLISH DEFINITION
HDMI	→	The cable that carries **both picture and sound** together — one cable does it all.
Input / Source	→	Tells the display **which device to show** — laptop, doc cam, room PC, etc.
Output / Display	→	**Where the image or sound appears** — the projector, flat panel, or speakers.
Signal	→	The **data traveling from your device to the display** — like water through a pipe.
AV Switcher	→	A **hub that lets multiple devices share one display** — switches between them on demand.
Control Panel	→	The **touchscreen or remote that runs the whole room** — power, volume, and input selection.

"These six terms will solve 90% of your confusion" — Master the vocabulary and you can diagnose almost any classroom AV problem.

QUICK WIN — GROUP ACTIVITY

3
MINUTE

Your **3-Point** Classroom AV Check

1. **Where is the INPUT / SOURCE selector?**
   Usually on the touchpanel, wall remote, or AV switcher
   Touchpanel / Remote

2. **Where is the VOLUME control?**
   Could be on the touchpanel, wall panel, or a dedicated amplifier knob
   Panel / Touchscreen

3. **Where is the DISPLAY POWER button?**
   Projector power button, flat panel remote, or touchpanel system button
   Remote / Touchpanel

Right now — look around the room and identify all three.

Tell the person next to you where they are.

Speaker note: 3 min group activity — pause the lecture. Walk the room as participants look around. Celebrate any who can immediately name all three.

MODULE 02

Audio Fundamentals

Why What They Hear Matters More Than What They See

KEY INSIG

Poor audio — not poor video — is the **#1 driver** of student disengagement.

75 minutes • Lecture + Demonstration

MODULE 02 — AUDIO FUNDAMENTALS

SIGNAL FLOW

Source → Destination

The Classroom **Audio Chain**

Follow the sound — every step in this chain is a potential failure point.

VOICE / MEDIA → **MICROPHONE** → **DSP / MIXER** → **AMPLIFIER** → **SPEAKERS** → **STUDENTS' EARS**

- **VOICE / MEDIA** — The source of sound
- **MICROPHONE** — Captures the sound
- **DSP / MIXER** — Processes and balances
- **AMPLIFIER** — Increases power
- **SPEAKERS** — Distributes sound
- **STUDENTS' EARS** — The destination

KEY INSIGHT **Troubleshooting audio = finding where this chain broke.**
Start at the source. Work forward until the break reveals itself.

ANSI S12.60 — **What the Standard Requires**

Acoustics Standard for Classrooms

BACKGROUND NOISE

Maximum Allowable Level

35 dB(A)

Occupied, unoccupied room

About as quiet as a very soft whisper — *most classrooms fail this standard* due to HVAC noise and building vibration.

SPEECH LEVEL

At Every Student's Ear

65–70 dB

Measured at the listening position

A normal raised speaking voice at 6–8 feet. Your mic and speaker system must deliver this level to every seat in the room.

REVERBERATION (RT60)

Maximum Echo Decay Time

0.6 sec

Time for sound to decay 60 dB

How long echoes persist after a sound stops — hard surfaces (concrete, glass, bare walls) make this significantly worse.

Why it matters: If your room fails these standards, your audio system must compensate — and often cannot fully make up the difference. Acoustic treatment is the first fix; amplification is second.

Microphone Types for Classroom Use

MODULE 2 · AUDIO FUNDAMENTALS

MIC TYPE	WHAT IT LOOKS LIKE	BEST USE	WHERE FOUND
Lavalier / Lapel	Tiny capsule mic that clips to the collar or lapel; thin cable runs to a bodypack transmitter	One-on-one teaching Lecture & movement	
Boundary	Flat disc or rectangle that lies on a table surface; hemispherical pickup captures all voices around it	Conferences & roundtables Group discussion	
Ceiling Array	Flush-mounted tile or pendant hung from ceiling; multiple elements form a wide pickup zone covering the whole room	Active classrooms High-movement teaching	
Handheld Dynamic	Traditional torpedo-shaped mic held in the hand; rugged cardioid capsule, wired or wireless	Presentations & Q&A Student response	
Gooseneck	Flexible metal-necked mic mounted on a podium base; bends to position capsule near the speaker's mouth	Podium lecturers Panel discussions	

Activity (7 min): Look around your classroom right now — which mic type do you have? Tell the person next to you. If you're unsure, that's your first troubleshooting task back at school.

MODULE 03

Video and **Displays**

Understanding What Hangs on Your Wall — and How It Works

KEY NUMBERS YOU'LI

- Room Depth ÷ 8
- Throw Ratio
- 6× Display Height

60 MINUTES

8 min

Display Types — **Know What You Have**

LED FLAT PANEL

✓ PROS

- Low maintenance — no consumable parts
- Vivid, high-contrast color in any light

✗ CONS

- Image size limited by panel size — not scalable

Best for rooms < 40 ft depth

STANDARD LAMP PROJECTOR

✓ PROS

- Affordable upfront cost
- Large image — scalable to room size

✗ CONS

- Lamp burns out — replacement $200–$400
- Requires dim lighting for best image

Medium to large rooms

LASER PROJECTOR

✓ PROS

- No lamp — 20,000+ hour lifespan
- Works well in ambient (lit) classrooms

✗ CONS

- Higher upfront purchase cost

Any room size, any lighting condition

INTERACTIVE FLAT PANEL

✓ PROS

- Touch-enabled — fully collaborative
- Replaces whiteboard + display in one unit
- Ideal for active learning environments

✗ CONS

- Limited to room depth < 40 ft

Best for active learning / small-medium rooms

Identify yours: Look at your classroom display right now — which type is it? Knowing this determines your maintenance responsibilities and lighting needs.

Speaker note: 8 min — Ask participants to identify their classroom display ty

MODULE 3 — VIDEO & DISPLAYS

Three Numbers Every Teacher Needs

01 FORMULA ONE

Room Depth ÷ 8
= Minimum Display Height

EXAMPLE
32 ft room ÷ 8 = **4 ft (48 in)** min. display height

If your display is smaller than this, back-row students cannot read projected content — no matter how large the font appears to you at the front.

Back-row readability risk

02 FORMULA TWO

Throw Ratio = Distance ÷ Width
Projector Distance ÷ Image Width

EXAMPLE
A **2.0** throw ratio needs **20 ft** distance to make a **10 ft** wide image

Wrong projector placement causes blurry, undersized, or keystone-distorted images. This ratio tells you whether a projector physically fits your room geometry — your AV team uses it to spec the right unit.

Critical for projector rooms

03 FORMULA THREE

Max Viewing Distance = 6× Height
6 × Display Height = Max Seat Depth

EXAMPLE
4 ft display → max seating depth of **24 ft**

Seats beyond this distance cannot reliably read 18pt text. If your room has seats past the threshold, your display is undersized — report it using the formula so IT can take action.

18pt text legibility limit

You don't calculate these — your AV designer does. But knowing them helps you **report problems accurately** and advocate for the right fix when something looks wrong in your classroom.

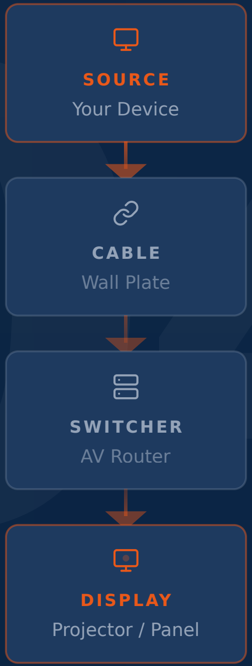
MODULE 04

Signal Flow & Connectivity

Tracing the Path From Your Device to the Display

Every AV problem lives somewhere on this chain. **Find where it broke.**

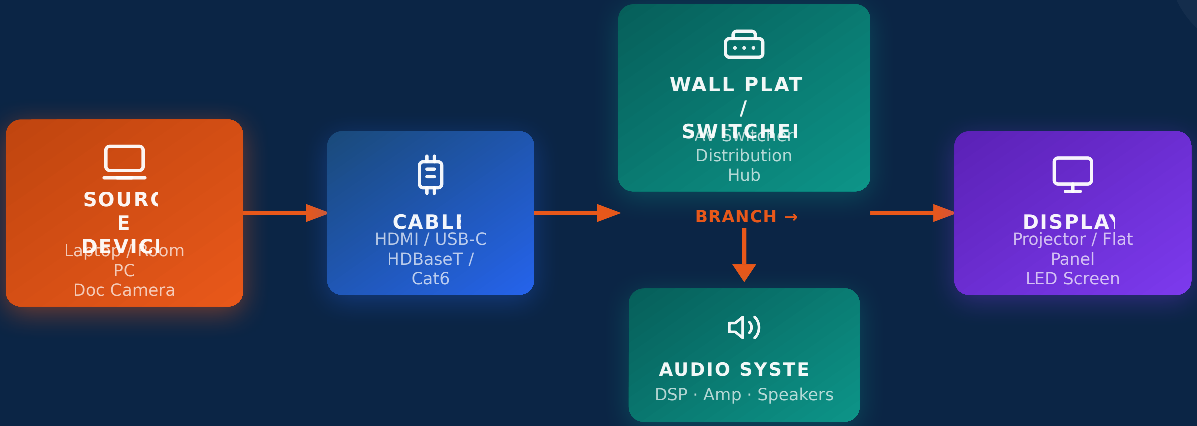
60 minutes | LECTURE + DEMONSTRATION



MODULE 4 Signal Flow & Connectivity

TROUBLESHOOTING MAP
6 min — Lecture

The **Signal Flow** Chain



Plays the content | Carries the signal | Routes the signal | Shows the content

Every troubleshooting session starts here. Ask: **WHERE did the signal break?** · This diagram is your troubleshooting map — we'll use it all day tor

QUICK CHECK: • **Is source ON?** → • **Is cable connected?** → • **Is input correct?** → • **Is display on?**

MODULE 4 — SIGNAL FLOW & CONNECTIVITY

Cable & Connector Quick Reference

DIGITAL ANALOG MULTI-SIGNAL

CONNECTOR	CARRIES	MAX LENGTH	WHERE FOUND
HDMI DIGITAL	Audio + Video — both signals in one cable		Projector, flat panel display, room PC
USB-C / Thunderbolt MULTI	Audio + Video + Data + Power	6 ft	Modern laptops, docking stations
HDBaseT / Cat6 DIGITAL	HDMI signal over network cable	328 ft (100m)	Long classroom runs, wall plates
VGA ANALOG	Video only — legacy, no audio	50 ft	Older classroom equipment
3.5mm Audio ANALOG	Audio only	15 ft	Speaker / headphone connection
XLR ANALOG	Balanced professional audio	200+ ft	Microphones to DSP / mixer

⚠ **Never mix digital (HDMI, USB-C) and analog (VGA, 3.5mm) signal paths without an active converter.**
Passive adapters will degrade or drop the signal entirely.

HDMI × VGA without converter

⏱ 8 min — Speaker Note: Walk through each row; ask participants to identify cable types in their own r

MODULE 4 · SIGNAL FLOW & CONNECTIVITY

The Educator's Proactive AV Tech Kit

Your Starter Kit

5 min

1 — ESSENTIAL CABLES

- [ ] HDMI cable — 10 ft
- [ ] USB-C to USB-A cable
- [ ] IEC power cord — backup for display / projector

Tip: Zip-tie cables and label both ends.

2 — ADAPTERS & DONGLES

- [ ] USB-C to HDMI adapter
- [ ] HDMI to VGA adapter (legacy rooms only)

Do not mix digital (HDMI, USB-C) and analog (VGA, 3.5mm) signal paths without an active converter.

3 — BATTERIES

- [ ] AA batteries — 4 or more
- [ ] AAA batteries — 4 or more

30%
of apparent system failures are dead batteries — **check remotes first.**

4 — REFERENCE

- [ ] This handbook — Modules 1–6
- [ ] IT Help Desk number posted at AV station

Fill in your own kit checklist in the handbook now — you'll reference it tomorrow.

Total kit cost: approximately $40-60. Prevents **hours of lost instructional time** per semester. Fill in your checklist in the handbook now

MODULE 05

60 minutes — Introduction Only

Networking Basics

What You Need to Know — and What You Don't

IP Addresses & Device IDs | Bandwidth for HD Video | Separate AV Networks | Firmware Updates | Wired vs. Wi-Fi

IP addresses, VLANs, and QoS are IT's job. Today you're learning the **5 things** that actually affect your classroom.

5 Things Every Teacher Should Know About AV Networking

8 MIN · DISCUSSION

1. **AV devices have IP addresses — if one changes after a reboot, the device goes "offline."**
   Tell IT the exact device name and location. Never just say "the projector isn't working."
   IP ADDRESS

2. **HD video calls need 3–8 Mbps upload AND download — symmetrically.**
   A shared Wi-Fi network with 30 students may not reliably deliver this. Request a dedicated connection.
   BANDWIDTH

3. **AV and student devices are often on separate networks on purpose.**
   This is a security feature — not a mistake. Report it to IT if it blocks legitimate classroom functions.
   NETWORK SEG.

4. **Firmware updates fix many chronic, unexplained problems.**
   Ask IT when a misbehaving device was last updated. Outdated firmware is a silent troublemaker.
   FIRMWARE

5. **Wired Ethernet is always better than Wi-Fi for AV — treat wireless as backup only.**
   If your AV station has a wired port, plug in. Reserve Wi-Fi for devices with no other option.
   WIRED FIRST

Pause here — questions welcome before moving to Module 6.

MODULE 06

Accessibility and Equity

Every AV Decision Is an Access Decision

KEY PRINCIPLE

Accessible AV benefits **every learner** — not just those with documented disabilities.

45 minutes • Lecture + Discussion • Day 1 · Module 6 of 6

7 MIN

Minimum Accessibility Practices — Your Checklist

CAPTIONS & AUDIO

- **Enable closed captions** on every video by default
- **Use room microphone** when available
- **Minimum 24pt body text / 36pt key points** for projected content
- **Never show an uncaptioned video**
  unless no captioned version exists

VISUAL & PHYSICAL ACCESS

- **Avoid red-green color distinctions alone**
  8% of males have color vision deficiency
- **Minimum 4.5:1 contrast ratio** for all displayed text
- **Know where assistive listening system** receivers are stored
- **Report ADA/accessibility AV failures immediately**
  These are civil rights compliance issues

Captions help: ELL students, auditory processing, noisy rooms, and study review. · **Make captions default — always on.**

ADA **Quick Reference** for AV

CIVIL RIGHTS COMPLIANCE

REQUIREMENT	STANDARD	WHAT TEACHERS DO
Background Noise	**≤ 35 dB(A)** ANSI S12.60	**Close doors** and windows during instruction; report **chronic HVAC noise** to facilities with a specific description (constant hum, intermittent banging, etc.).
Assistive Listening Systems	**ADA Required** All amplified rooms	**Know where receivers are stored** and how to enable the system. Offer receivers proactively — do not wait for students to self-identify a need.
Captions	**FCC / ADA** Broadcast content	**Turn on captions by default** — every video, every time. Never skip captions. Benefits ELL students, noisy rooms, and study review equally.
Control Panel Height	**Max 48 in. AFF** ADA Standards §308	**Report inaccessible controls immediately.** If a panel exceeds 48 inches above finished floor, document location and submit a facilities ticket — this is a code violation.

KEEP TEACHING

If your room fails **any** of these standards, document it and report it. **You are the first line of accessibility compliance awareness** — your report triggers the fixes that protect every learner.

DAY 1 WRAP-UP

⏱ 5 min · Distribute kits if applicable

Day 1 Complete — **Great Work!**

WHAT YOU LEARNED TODAY

- The **4 layers** of every AV system and who controls each
- How **audio & video signals** travel from source to display
- **Cable types** and what each connector carries
- Your **Proactive AV Tech Kit** (~$40–60 total)
- **Accessibility** as a core AV requirement — not optional

Module 1 · Module 2 · Module 3 · Module 4 · Module 5 · Module 6

WHAT TO DO TONIGHT

- [ ] Review your handbook Modules 1–6 — focus on terms you want to remember
- [ ] Complete your Proactive Kit Checklist in the handbook
- [ ] Identify your classroom mic type from today's reference table
- [ ] Look up your IT Help Desk number and add it to the AV station card

Takes about 20 minutes — worth every one of them.

WHAT'S COMING TOMORROW

FULL DAY 2 FOCUS: TROUBLESHOOTING

No Image scenarios — step-by-step diagnosis

No Audio scenarios — step-by-step diagnosis

Video conferencing failures & live fixes

Escalation framework — when to call IT and what to say

Capstone scenario stations — hands-on practice

Day 2 of 2 · 7 Hours · Bring your kit

Tomorrow you will fix things. Tonight, get some rest.